

SOHAM DHEEMATE

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Full-stack and AI engineer building production RAG pipelines, multi-agent systems, and real-time web applications. Designed an enterprise MS Teams bot architecture (8 sub-agents, 100+ API tools) approved by senior management at KPIT. Skilled in LangChain, Node.js, React, Angular, Python, and .NET across applied AI and full-stack projects.

WORK EXPERIENCE

Software Engineer | **KPIT Technologies** Mar 2026 – Present · Pune, India

- Designed and leading a multi-tier MS Teams bot with an intent classifier routing to 8 specialised sub-agents exposing 100+ API tools — architecture proposed independently and approved by senior management after a live 2-agent demo
- Built a full RBAC system for the VOC product with complete UI and backend gating using Angular, .NET, and SQL — includes roles list, user management, role-permission mapping, and an email configuration module to trigger notifications on permission and role change events; account-level admins and PMs can grant additional access without overriding admin-level permissions; architecture designed to roll out across other KPIT products

Software Engineer Intern | **KPIT Technologies** Nov 2024 – Feb 2026 · Pune, India

- Built a RAG POC supporting all major file types (PDF, Excel, JSON, TXT) to make internal project documentation searchable via natural language
- Engineered a LangChain agent with dual tools — a RAG retriever and a SQL tool that converts natural language to SQL, executes against the live database, and returns answers in plain English — resolving precision failures on structurally similar data where pure RAG was insufficient
- Replaced HTTP polling with WebSockets in the VIP global display system, eliminating redundant request cycles and stale cache across worldwide display screens

SDE Intern | **Ravvio** Jan 2025 – Jun 2025 · Remote

- Built responsive React.js chat UI with reusable components (chat bubbles, typing indicators) using Tailwind CSS and Context API, supporting seamless real-time conversations across desktop and mobile
- Developed and integrated RESTful APIs in Node.js/Express.js for message delivery and retrieval, improving chat reliability and overall user experience
- Implemented centralised error handling and input validation across all backend routes, ensuring consistent, fault-tolerant API behaviour in production

PROJECTS

Study Abroad Buddy [Live](#)

Node.js · React · LangChain LCEL · Pinecone · Socket.io · Redis · MongoDB · JWT · Vercel · Render

- Built a full-stack study-abroad guide app with 5 LangChain LCEL agents (Visa, Health, Culture, Housing, General) and an intent classifier routing user queries to the correct specialist agent
- Implemented a RAG pipeline using Pinecone (3 namespaces) with OpenAI embeddings; responses grounded in curated source documents
- Delivered real-time token-by-token streaming via Socket.io with JWT auth on the socket handshake; Upstash Redis caching (1-hour TTL) eliminates redundant OpenAI calls on repeated queries
- Shipped AI-generated personalised checklists, guide pages with MongoDB caching, persistent UI state, and a 3-step onboarding wizard — deployed live on Vercel + Render

Wardrobe.AI ★ **Best Project Award — AIA Dept.**

Python · FastAPI · GPT-4o-mini · OpenAI Embeddings · ChromaDB · LangChain · MediaPipe Pose · React

- Combined vision AI, RAG, and AR: GPT-4o-mini reads uploaded clothing images, generates style suggestions, and retrieves the best-matching item via cosine similarity on a ChromaDB vector store
- Built a MultiVectorRetriever architecture separating the embedding store from the image store, enabling text-based similarity search while returning actual product images

- Delivered an in-browser AR try-on using MediaPipe Pose to detect 33 body landmarks per frame and overlay garments scaled to the user's torso in real time

Personalized Cricket Player Analysis

React · Python · K-Means Clustering · Node.js — IEEE Published, MIT ADT International Conference 2024

- Built a web-based real-time scoring platform to digitise grassroots cricket data, capturing ball-by-ball entries across custom attributes — shot quadrant (8 zones), pitch position (20 subareas), bowler type, shot type, and runs — converting paper-based match records into structured CSV data stored in a database
- Applied K-Means clustering to ball-by-ball player data forming 6 distinct performance clusters; analysis revealed actionable patterns — e.g. batsmen facing shorter-pitched deliveries scored predominantly on the leg side while fuller pitches produced offside play — providing strategic insights for bowlers and coaches

SKILLS

Languages: Python · JavaScript · TypeScript · C# · Java · C++

Full-Stack: React · Angular · Node.js · Express · .NET · Tailwind CSS · Socket.io · REST APIs · JWT

AI / LLM: LangChain LCEL · LangChain Agents · OpenAI API · RAG · Pinecone · ChromaDB · Multi-Agent Systems

Databases: SQL · MongoDB · Mongoose · Redis

Tools: Git · Postman · Vercel · Render

EDUCATION

B.Tech in CSE — Specialisation in Artificial Intelligence 2021 – 2025

MIT ADT University, Pune · CGPA: 7.69

HSC (Class XII) 2019 – 2021

Arham High School and Junior College, Pune · Aggregate: 74%

ICSE (Class X) 2018 – 2019

The Bishops School, Undri, Pune · Aggregate: 91%

CERTIFICATIONS

- Mastering Data Structures & Algorithms using C and C++ — Udemy
- Cloud Computing Applications: Cloud Systems and Infrastructure — Coursera (UIUC)
- Node.js, Express, MongoDB & More: The Complete Bootcamp — Udemy

ACHIEVEMENTS & EXTRACURRICULARS

- Best Project Award — AIA Department, MIT ADT University, for capstone project Wardrobe.AI
- IEEE publication — MIT ADT International Conference 2024 (Personalized Cricket Player Analysis)
- Team Lead, Institution's Innovation Council (IIC) — MIT ADT University
- Played football at university level and in the Pune District Football Association (PDFA) league